Recognition Task-Based Detection Score

A Task-Oriented Evaluation Metric for Infrared Image Colorization

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Hao Wang, Jiaming Cai, Yao Hu*, Chenglong Zhang, Qun Hao

Presentation Outline

Background & Methods

1. Introduction & Motivation
2. RDS Metric Definition
3. Category Division Strategy
4. Datasets Construction
5. Training Strategy

Experiments & Results

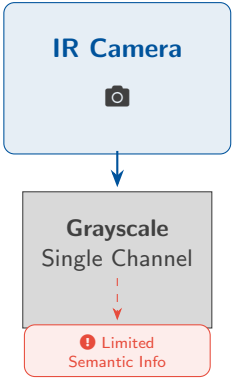
6. Effectiveness Validation
7. Position Robustness
8. Fine-Grained Interpretability
9. Task Adjustability
10. Limitations & Conclusions

Intro → Methods → Experiments → Results → Conclusion

Infrared Imaging Technology

Key Characteristics

- **Independent of visible light** illumination



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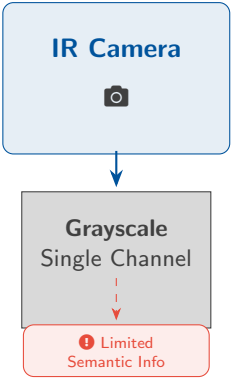
- **Stable perception** in complex environments



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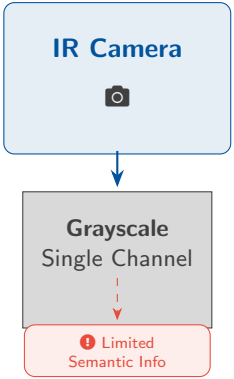
- **Applications:**
  - Night vision surveillance
  - Autonomous driving
  - Military reconnaissance



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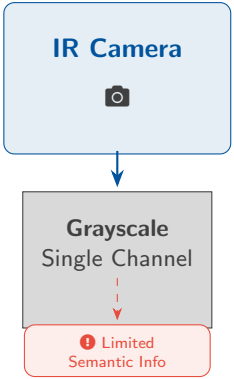
Limitations

- Single-channel grayscale images
- Limited degrees of freedom in color space
- Insufficient semantic information
- Constrained detection/recognition performance



Infrared Image Colorization

Image-to-Image Translation

Converting infrared grayscale images to color images through deep learning methods

CNN-based
Multi-scale CNN
S-shape Network
U-Net
2016–2020

Evolution →

GAN-based
CycleGAN
Pix2pix
TIC-CGAN
2017–Present

Trend →

Emerging
Diffusion Models
Vision Transformers
2023–Future

Benefits

- Enhanced visual presentation
- Improved semantic expression
- Better human interpretation

Challenge

How to evaluate colorization quality objectively?

Existing Evaluation Methods

No-Reference Metrics

- Compare before/after colorization

Full-Reference Metrics

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- **Example:** APCE (Average Precision of Canny Edges)

Full-Reference Metrics

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Full-Reference Metrics

- Compare with reference images
- **Examples:**
  - SSIM – Structural Similarity
  - PSNR – Peak Signal-to-Noise Ratio
  - FID – Fréchet Inception Distance

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- ✗ Focus on pixel-level similarity
- ✗ Ignore downstream task support

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Key Problem:
Existing metrics fail to reflect the **usability of colorization results in practical tasks**

The Need for Task-Oriented Evaluation

Key Insight

Colorization quality should be assessed from its **support capability for downstream tasks**

Position
Robustness

Tolerate registration errors
maintain stability

Fine-Grained
Interpretability

Category-level analysis
identify weaknesses

Task
Adjustability

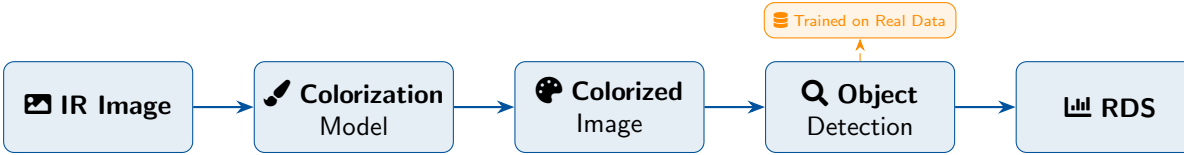
Flexible evaluation criteria
match task objectives

Solution: Recognition Task-Based Detection Score (RDS)

RDS: Recognition Task-Based Detection Score

Core Idea

Use the **recognition accuracy of object detection models** on colorized images as a proxy indicator for colorization quality



Position Robustness

IoU-based matching
Tolerates position errors

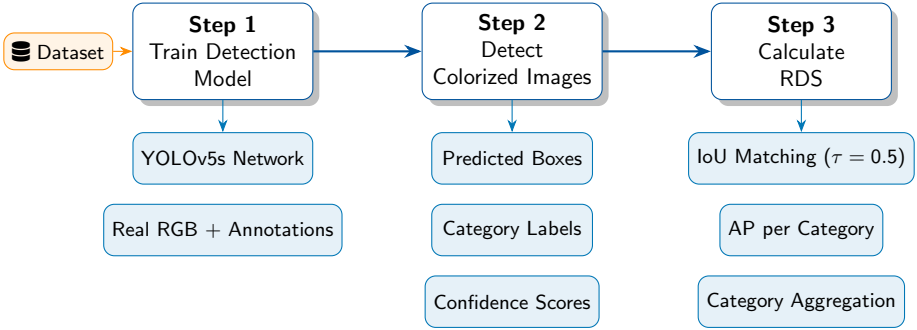
Fine-Grained

Category-level AP
Per-class analysis

Task Adjustable

Flexible categories
Match objectives

RDS Computation Pipeline

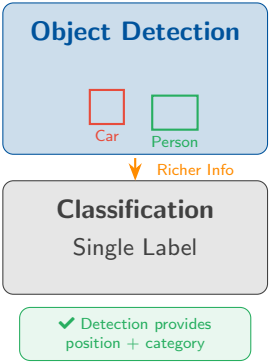


$$\text{RDS} = \frac{1}{N} \sum_{i=1}^{N} \text{AP}_i$$

Why Object Detection?

? Task Selection Rationale

- **vs. Image Classification:**
  - Detection requires **spatial localization**
  - More sensitive to local color, texture, structure



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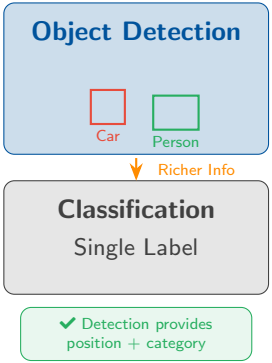
- **Semantic Preservation:**
  - Recognition accuracy = semantic structure quality
  - Direct link to downstream usability



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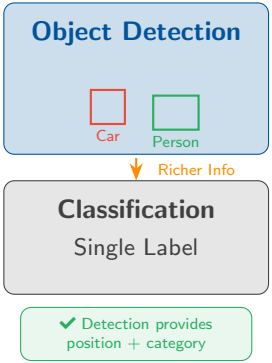
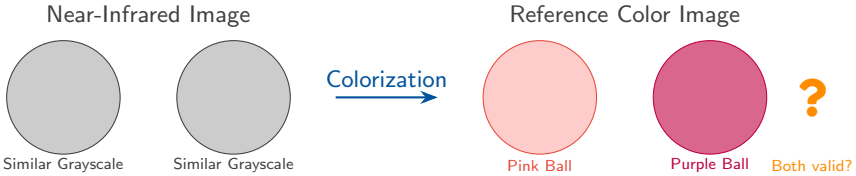
- **Practical Advantages:**
  - Category decomposition capability
  - Spatial matching mechanism (IoU)
  - Mature, efficient models available



Single-Shape-Multiple-Color Problem

The Challenge

The same grayscale shape can correspond to **multiple possible colors**



Seen Scenes

- Color can be **learned and memorized**
- Metric should emphasize **color restoration**

Unseen Scenes

- Color prediction is **uncertain**
- Metric should emphasize **structure preservation**

Category Division Strategy

Scene-Specific Mode

- Retain shape + color
- $\mathcal{C}_{\text{memory}} = \mathcal{C}_{\text{orig}}$
- Seen scenes
- Emphasize **color restoration**

Scene Generalization Mode

- Retain shape only
- $\mathcal{C}_{\text{general}} = \{s_i | s_i \in S\}$
- Unseen scenes
- Emphasize **structure preservation**

Original: $\mathcal{C}_{\text{orig}} = \{(s_i, c_i) | s_i \in S, c_i \in C\}$
Categories defined by **shape** s_i and **color** c_i

Dataset Construction: Overview

NIR–RGB Dataset
Indoor scenes
Controlled conditions
High registration accuracy

Multispectral | 6 Objects

FLIR-5C Dataset
Outdoor scenes
Traffic scenarios
Complex backgrounds

Thermal | 5 Categories

Purpose: Evaluate model learning, memory, and generalization capabilities

NIR–RGB Dataset Details

Equipment

- **Camera:** FSFE-1600D-10GE
- **Type:** Multispectral area-scan
- **Method:** Prism light combination
- **Resolution:** 640×480

Data Type	Images	Targets	Scenes
Training set	1,443	4,134	138
Seen Test Set	361	1,025	–
Unseen Test Set	205	651	33

Table: NIR–RGB Dataset Statistics

Object Categories

1. Black jar
2. Red cube
3. Brown cup
4. White cup
5. Pink ball

Key Feature

Hardware-level pixel alignment ensures **high registration accuracy**

FLIR-5C Dataset Details

Source

- Based on publicly available **FLIR thermal dataset**
- Category filtering and grayscale processing
- Simulates infrared-color image pairs

Five Categories

- Person
- Car
- Bus
- Traffic light
- Traffic sign

Data Type	Images	Targets	Scenes
Training set	8,403	129,172	132
Seen Test Set	943	14,742	–
Unseen Test Set	1,033	15,565	17

Table: FLIR-5C Dataset Statistics

Application

Real-world traffic scene evaluation for **generalization testing**

Training Strategy

Detection Model Training

YOLOv5s – Lightweight single-stage detector balancing accuracy and efficiency

Training Configuration

Parameter	Value
Resolution	640×480
Epochs	100
Optimizer	Adam
Initial LR	1×10^{-3}
Final LR	1×10^{-6}
LR Schedule	Cosine annealing

Platform

- **OS:** Ubuntu
- **Framework:** PyTorch 1.11.0
- **CPU:** Intel Xeon Platinum 8255C
- **Memory:** 43 GB
- **GPU:** NVIDIA RTX 3090 (24 GB)

Experimental Setup

Colorization Models

1. **CycleGAN**
   Unpaired image translation
2. **Pix2pix**
   Conditional GAN
3. **TIC-CGAN**
   Improved with structural constraints

Comparison Metrics

- **SSIM** – Structural Similarity
- **PSNR** – Peak Signal-to-Noise Ratio
- **FID** – Fréchet Inception Distance
- **RDS** – Proposed metric

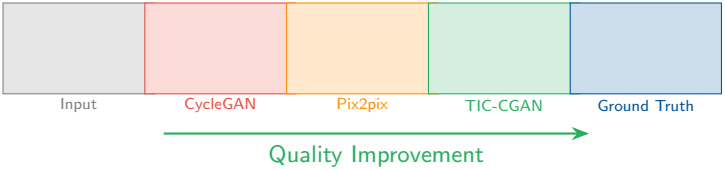
Training: 500 epochs (NIR-RGB) / 100 epochs (FLIR-5C)
Resolution: 640×480 — **Optimizer:** Adam — **LR:** $10^{-3} \rightarrow 10^{-6}$

Effectiveness Validation: Visual Results

Subjective Visual Comparison

Quality ranking consistent across both datasets

FLIR-5C Dataset



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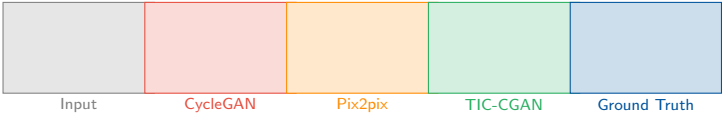


Effectiveness Validation: Quantitative Results

Model	FLIR-5C Seen Test				FLIR-5C Unseen Test			
	SSIM	PSNR	FID	**RDS**	SSIM	PSNR	FID	**RDS**
CycleGAN	0.848	26.19	75.35	31.13%	0.833	25.73	77.40	30.94%
Pix2pix	0.899	28.81	57.82	34.60%	0.881	27.73	66.16	33.78%
TIC-CGAN	**0.944**	**31.31**	**38.70**	**38.39%**	**0.937**	**30.84**	**40.80**	**39.87%**

Key Finding

RDS ranking matches both:

- Subjective visual evaluation (TIC-CGAN > Pix2pix > CycleGAN)
- Traditional metrics (SSIM, PSNR, FID)

Effectiveness Validation: NIR–RGB Results

Model	NIR–RGB Seen Test				NIR–RGB Unseen Test			
	SSIM	PSNR	FID	**RDS**	SSIM	PSNR	FID	**RDS**
CycleGAN	0.788	17.01	136.2	57.39%	0.715	14.89	146.2	50.81%
Pix2pix	0.892	**26.03**	71.97	91.38%	0.784	17.55	125.0	74.06%
TIC-CGAN	**0.911**	25.79	**31.26**	**95.71%**	**0.816**	**18.08**	**74.35**	**76.05%**

Important Interpretation

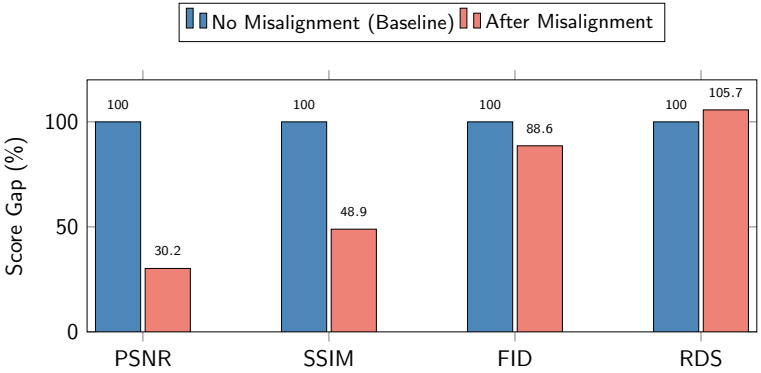
On NIR–RGB seen scenes, **Pix2pix has slightly higher PSNR** than TIC-CGAN (26.03 vs. 25.79), but TIC-CGAN still wins on **SSIM, FID, and RDS**.

This is a good point to mention in presentation: **pixel fidelity and task utility are not always identical.**

Position Robustness Validation

↔ Experiment Design

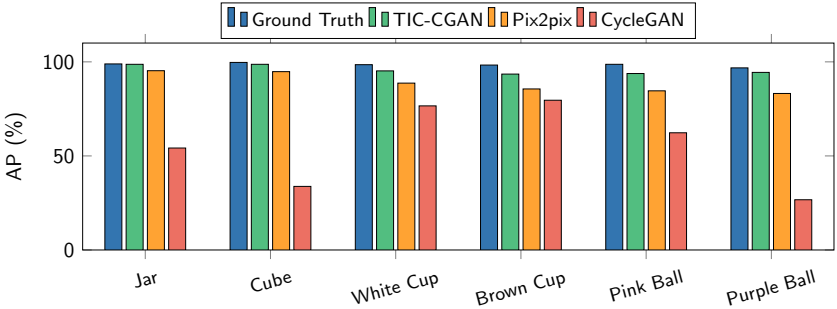
Artificial misalignment: Random 1–5 pixel shift in one direction (up/down/left/right)



Fine-Grained Interpretability

Category-Level AP Analysis

RDS reveals model performance differences across different object categories



Key Insights

CycleGAN weakness: AP only 26.7% on "Purple Ball" – indicates color restoration failure

Task Adjustability: Category Strategy Comparison

Two Evaluation Modes

Scene-Specific Mode (6 categories): Emphasizes color restoration
Scene Generalization Mode (4 categories): Emphasizes structure preservation

Scene-Specific Mode

CycleGAN	50.81%
Pix2pix	74.06%
TIC-CGAN	76.05%

Category Merge →

Scene Generalization Mode

CycleGAN	73.39%
Pix2pix	93.70%
TIC-CGAN	**96.45%**

↑ TIC-CGAN: 76.05% → 96.45%
20.4 percentage point improvement!

Conclusion

Flexible category definition avoids penalizing reasonable colorization in unseen scenes

RDS: Three Key Characteristics

Position Robustness

IoU-based matching

Tolerates registration errors

5.7% improvement

vs. 11–70% degradation

Fine-Grained Interpretability

Category-level AP

Reveals model weaknesses

26.7% AP

on weak category

Task Adjustability

Flexible category division

Matches task objectives

76% → 96%

RDS improvement

RDS provides reliable evaluation under imperfect conditions,
diagnostic insights for improvement, and
adaptable criteria for diverse applications

Comparative Analysis

Capability	SSIM	PSNR	FID	RDS
Pixel-level Similarity	✓	✓	✗	✗
Feature Distribution	✗	✗	✓	✗
Task-Oriented	✗	✗	✗	✓
Position Robustness	✗	✗	●	✓
Category-Level Analysis	✗	✗	✗	✓
Task Adjustability	✗	✗	✗	✓

Table: Capability Comparison of Evaluation Metrics

Key Takeaway

RDS is the **only metric** that provides task-oriented evaluation with position robustness, fine-grained analysis, and task adjustability

Conclusions

Main Contributions

1. **Novel Task-Oriented Metric:** RDS bridges the gap between colorization quality evaluation and downstream task performance

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2. **Three Key Design Characteristics:**
   - Position robustness via IoU-based matching
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2. **Three Key Design Characteristics:**
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   - Task adjustability via flexible category division
3. **Comprehensive Validation:** Demonstrated effectiveness, robustness, interpretability, and adjustability through systematic experiments

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   - Task adjustability via flexible category division
3. **Comprehensive Validation:** Demonstrated effectiveness, robustness, interpretability, and adjustability through systematic experiments

Quantitative Highlights

- **Stability:** only a +5.7% fluctuation under misalignment vs. up to 69.8% degradation for traditional metrics
- **Diagnostic:** Reveals per-category weaknesses (e.g., 26.7% AP on specific category)

Future Work

Broader Applications

- Night vision enhancement
- Remote sensing false color
- Medical imaging
- Underwater imaging

Extended Tasks

- Semantic segmentation
- Action recognition
- Depth estimation
- Multi-task learning

Research Directions

1. **Architecture Validation:** Test RDS consistency across YOLOv8, Faster R-CNN, etc.
2. **Modern Methods:** Apply to Diffusion Models and Vision Transformers
3. **Human Correlation:** Spearman correlation analysis with subjective ratings

Limitations of This Paper

Methodological Limits

- **Single proxy detector:** validation is based only on YOLOv5s in the paper
- **Single downstream task:** only object detection is used, not segmentation or retrieval
- **No formal human-study correlation:** consistency is argued qualitatively, not with a large-scale subjective study

Experimental Limits

- **Limited model diversity:** only CycleGAN, Pix2pix, and TIC-CGAN are compared
- **Dataset availability:** the custom NIR–RGB data are not public in the paper
- **Possible detector bias:** if detector training is weak, RDS may reflect detector weakness too

Balanced Take

RDS is strong as a task-oriented metric, but it is not a complete replacement for perceptual metrics

Key Takeaways

RDS: From Pixel Similarity to Task Utility

A paradigm shift in infrared image colorization evaluation

Robust
Works under imperfect conditions

Diagnostic
Reveals specific model weaknesses

Adaptable
Matches diverse task objectives

"Evaluation metrics are key to driving progress in machine learning
– Jayasumana et al., CVPR 2024

Thank You!

Questions & Discussion

huy08@bit.edu.cn

Beijing Institute of Technology

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